

What Machine Learning Can Learn from Foresight

A Human-Centered Approach

For machine learning-based forecast efforts to succeed, they must embrace lessons from corporate foresight to address human and organizational challenges.

Christian Crews

OVERVIEW: Machine learning applications in business that return forecasts or predictions of future market or consumer behavior must pay attention to nontechnical aspects of how those forecasts are created and used by leaders. Machine learning projects can generate better forecasts that have greater effect by embracing key methods developed through almost 50 years of corporate foresight practice to improve the adoption and use of forecasts in organizations.

KEYWORDS: Foresight; Machine learning; Artificial intelligence; Forecasting

In a recent O'Reilly survey on the state of machine learning adoption, 54 percent of respondents in North America said they were early adopters (36 percent) or sophisticated users (18 percent) of machine learning (Lorica and Nathan 2018). Applications of machine learning in business have grown significantly; machine learning patents grew at a compound annual rate of 34 percent between 2013 and 2017, and IDC forecasts that spending on AI and machine learning will grow from \$12 billion in 2017 to more than \$57 billion by 2021. Deloitte predicts that the number of machine learning pilots and implementations by businesses will double from 2017 to 2018 and double again by 2020 (Columbus 2018).

One of the major uses of machine learning in business is prediction. Salesforce is using its Einstein system to help users find new leads based on historical data; Netflix applies AI via an algorithm that uses viewing history to make remarkably nuanced recommendations for members of what to watch next; Amazon is experimenting with machine learning to predict which applicants will best fill open positions; and Arena is using it to predict which nurses in an application pool are the best fit for a hospital's culture and will stay the longest.

Christian Crews has more than 20 years of experience applying foresight to innovation and strategy for large corporations, nonprofits, and government agencies. He designed and implemented the IRI2038 project for IRI in 2012–2013 and is a leading practitioner and innovator in scenario planning methods. Christian is the Foresight, Discovery and Ideation practice lead for Kalypso, and a founding member and previous board member of the Association of Professional Futurists. He holds an MS in Studies of the Future from the University of Houston–Clear Lake and a BA in English from the College of William and Mary. christian@andspaceconsulting.com

DOI: 10.1080/08956308.2019.1541725

Copyright © 2019, Innovation Research Interchange.
Published by Taylor & Francis. All rights reserved.

While these efforts are getting better every day in solving the technical challenges of applying machine learning to forecasting, the process of forecasting also faces some unique challenges, challenges that are not scientific or technical in nature but human and organizational. Corporate foresight practitioners, in engaging corporate leaders in thinking about the future, have developed methods to navigate those barriers; applications of machine learning to forecasting could benefit from that learning.

Improving the ability of AI and machine learning to shape business decisions will require applying some lessons from almost 50 years of corporate foresight. Those lessons can help machine learning algorithms and practitioners address three core problems that plague any forecasting effort: disbelief in the forecast, lack of strategic context, and delegation of foresight thinking.

Disbelief in the Forecast

Forecasts have been disbelieved since the time of Cassandra. Corporate foresight practitioners have faced this truth as well, and those seeking to apply machine learning should anticipate it. The parable of Cassandra provides some insight into how to improve the impact of forecasts in organizations that might be resistant to accepting them.

Cassandra is a character from Greek myth, gifted with the ability to predict the future and cursed to never be believed. Like most myths, this story is meant to teach us something about human nature—in this case, how people react to foresight. Humans develop strong mental models of the future that they use to make decisions. Rather than examine new evidence and revise those mental models, most humans either bend new information to match their expectations or,

if the information can't be made to fit into their model, ignore it completely (Gonzales 2004). We know that the most implausible things can happen and the lead-up to these surprising events can often be seen in hindsight, but the signals are often ignored until the event occurs because they don't make sense in the context of the dominant mental model of the future. Even when the signals are read as they emerge, predicting an event ahead of time does nothing to improve the ability of a person, a company, or a society to accept the prediction, interpret its implications, and take meaningful action in response. People must be involved in the process of generating the prediction—in the foresight process—if they are to move beyond their mental models and accept alternative visions of the future (Crews 2015).

Some companies have learned this lesson. Royal Dutch Shell is famous for using scenario planning in 1972 to anticipate the oil crisis that began the following year and make adjustments to capitalize on changes to the global oil market that the leadership team believed were coming. But the team was presented with almost the same scenarios in 1971 and was not persuaded to change course (Kleiner 2008). Pepsico's Advanced Research and Long Term Research groups used an outside team to develop powerful scenarios in 2008, but those insights did not begin to shape the research portfolio until the company kicked off a different initiative in 2010 (Farrington, Henderson, and Crews 2012). What was the difference for these companies? In both cases, the later effort engaged leaders directly in the process of creating those alternative views of the future.

One lesson to take from these stories is that in foresight, the process is as important as, or perhaps even more important than, the answer. Machine learning faces a particularly difficult problem in engaging humans in its foresight processes. Companies can build the best machine learning algorithms in the world and have them accurately predict all sorts of things important to the companies' futures—how consumers will react to new flavors, how supplier pricing will change, when a product will move from niche to mainstream. But those companies will likely never act on those predictions because simply getting a forecast about the future, even one with the promise of technological accuracy, will not shift mental models enough to motivate action. People must be able to engage with the forecasts.

For machine learning, the method by which the models are produced is itself a barrier to engagement. For too many people, AIs are "mysterious machines" that produce answers a human can't interrogate because few, if any, can understand the process by which the algorithms arrive at those answers (Knight 2017). Corporate leaders can't engage directly with machine learning algorithms, nor can they interrogate how algorithms arrive at their answers. As a result, they may not see the reality at the heart of the forecast.

What they can do is be involved in the design and creation of the algorithm itself. Engaging leaders in what many would consider technical decisions—decisions about what data sources to include, how to weight different sources, or what kinds of visualization of answers to provide and how often

to provide them—can generate a feeling of ownership of the process and create buy-in for its results. In the context of forecasting and futures work, the building and operation of the algorithm should not be seen as purely technical problems for the engineers to solve; rather, these decisions should be treated as management problems, key parts of the foresight process that should involve the stakeholders who must believe in, and respond to, the results of that process.

Lack of Strategic Context

Corporate foresight practitioners know that a strong forecast that spurs action begins with a clear articulation of the strategic need for forecasting. The forecast has to be driven by a long-term strategic focus, not a short-term need for immediate results. Because an algorithm-driven AI can generate complex answers quickly, a machine learning-based forecast can easily find itself providing suggestions for long-term actions without a strategic context in which to understand those recommendations.

Most companies and their shareholders are highly focused on the next quarter's results. Shareholders expect the companies they invest in to deliver to current customers, to generate growth and dividends in the near term. That focus has certainly played a positive role in the productivity and economic growth of countries with free markets. It also locks companies into a set of short-term cycles, making it difficult to see far enough ahead to prepare for the future and leaving space only for incremental innovation. To compensate, many organizations have a few designated functions specifically tasked with looking out further, such as corporate strategy, R&D, and some consumer insight functions. Leaders delegate the long view to these specialist functions, allowing leadership to focus on the work of managing day-to-day activities. As a result, often, few employees know what a company's long-term strategy is. In one study, even in companies that had a well-articulated strategy, only 29 percent of employees could pick the correct one out of a lineup of the usual suspects (Leppert and Kotter 2013). If no one knows what the strategy is, or where it's coming from, no one can craft policies or projects that support it.

That's also a problem for corporate foresight work; without some shared, long-term vision or goal, it's impossible to know how to respond to future scenarios. As George Harrison sang in "Any Road," riffing on an exchange between Alice and the Cheshire Cat in *Alice in Wonderland*, "If you don't know where you're going, any road will take you there." Before a foresight project can create a framework for

The leadership team needs to make explicit the company's often unofficial strategic objectives and intent.

The results delivered by the algorithms need to be interpreted, and that interpretation is not possible without a well understood, widely known long-term strategy.

interpreting trends and forecasts, the leadership team needs to do the hard work of making explicit the company's often unofficial strategic objectives and intent (Farrington, Henderson, and Crews 2012). To accomplish this, corporate foresight projects often begin with extensive interviews designed to identify and articulate the company's current vision of the future (Van der Heijden 2005). This work allows the results of the foresight exercise to be considered in light of the strategy—it allows the foresight team, and company leaders, to determine which forecasts or trends are important for the company's future, and thus should be responded to, and which can be disregarded. Without this connection to the company's strategy, it becomes very difficult for the management team to interpret the results of foresight.

But generating that connection is time-consuming, and leaders' impulses, often, are to find a way to make foresight easier or quicker. Technology, with its presumed ability to provide neat solutions to messy human problems, is frequently the answer. Leaders who do not want to take the time to engage in a collaborative approach to developing foresight look to machine learning to solve their forecasting problems. Why engage in a lengthy process of internal alignment when they can invest in a technology that will just deliver the answers? Technology investments seem easy to understand and the return on investment easily calculated.

In reality, though, the alignment process can't be side-stepped, even in a technology-based foresight process. Machine learning algorithms are just as capable of returning important information about the future whose implications few people will understand as human foresight teams. The results delivered by the algorithms need to be interpreted, just as other kinds of foresight do, and that interpretation is not possible without a well understood, widely known long-term strategy. Getting the most out of a machine learning investment, like any investment in foresight, is not about the technology but about preparing the organization—taking the time to build a detailed, long-term strategy and drive clarity and alignment within the company so that when the algorithm identifies important information about the future, that information can be recognized and used to drive appropriate decisions.

Delegating the Foresight Function

A forecast is only as valuable as its inputs. Successful corporate foresight projects ensure that a diverse set of people, from across

corporate silos, are engaged throughout the process. Building diverse teams ensures a range of inputs and a wider perspective on the forecast's significance in the context of the corporation's current environment and strategic needs. Shifting machine learning functions to siloed parts of the company can reduce the focus and effectiveness of a forecast project.

Organizations have mental models of the future, just as individuals do (De Geus 1997). Over time, people working within the organization adapt to that organizational model, allowing it to shape their own; it gradually becomes more difficult for them to escape that frame and form divergent visions of the future. Royal Dutch Shell recognized the danger of such inward thinking; to escape it, the company cultivated a network of "extraordinary people" around the world who were doing interesting work in technology, history, social sciences, and politics (De Geus 1997). This network allowed the scenario planning team, and then the company's leaders, to escape their mental models enough to see the world through different lenses. In this wider context, trends that were seen as not important within the company might take on additional weight, while long-held internal beliefs about the future might be discounted.

When machine learning enters the picture, this problem can become more acute, exacerbated by the tendency of companies to relegate the building of machine learning algorithms to IT departments or external IT providers. One critical component of machine learning is *feature engineering*. In feature engineering, data scientists help the algorithm learn by categorizing data and adjusting the weighting of various types of data. It's the equivalent of teaching a child—just as teachers or parents pass down a bit of the experience of being a human to accelerate and optimize a child's learning experience, data scientists give the algorithm some analytical support to direct its learning. This kind of work is also referred to as a Humans-in-the-Loop (HITL) approach.

In the context of a machine learning-based forecast initiative, the algorithm is expected to learn about, and provide information about, customers, products, and markets. But, typically, few people who deal with those elements are included in the algorithm design. When feature engineering is performed within IT departments or by external service providers, without access to a network of people who can provide inputs to the algorithm, its learning may be directed in ways that ultimately can't produce the results the company needs. Inputs for these algorithm-training processes should be provided by those who know the most about the market, the company's customers, and its products, and which signals, sources, and categories should be privileged. Involving people who have both responsibility for sensing the market and experience in doing so can accelerate the algorithm's learning and increase the accuracy and relevance of its output.

Treating the design and management of the machine learning algorithm as an IT problem limits the valuable experience that the rest of the company can provide. Machine learning will succeed in creating real value for the organization only when the best teachers are brought in to help the AI learn.

ML DESIGN ELEMENTS



ORGANIZATION ENGAGEMENT

Quarterly reviews of milestones enhance algorithms.

Stakeholders decide on the cadence and level of reporting.

Feature engineering workshops using extraordinary people to broaden the identification of features and categories over time.

Collaborative focus question workshops select opportunity areas and data sources.

corporate foresight and its approaches to addressing the human and organizational challenges to the use of forecasting in organizations is critical to the success of machine learning projects. Even though the design and development of a machine learning algorithm is highly technical, it must be treated as a human management project that engages experts and leaders from across the company. Engaging the full organization, and especially its leaders, will improve the results produced by machine learning, help leaders understand the importance of the discoveries the

algorithm delivers, and increase their motivation to act on those findings.

FIGURE 1. A human-centered approach to machine learning

Developing a Human-Centered Approach to Machine Learning–Driven Foresight

To ensure the success of an investment in machine learning for forecasting, three lessons can be adopted from the foresight discipline:

- *Remember that the process is as important as the answer.* Involve and engage leaders in the organization in the design and development of the algorithm to ensure that they own, and are prepared to act on, the results.
- *Create a strategic framework.* The results of machine learning should be interpreted in the context of their likely impact on the company; the strategic framework provides a frame of reference for understanding that impact and its effect on the company's strategic intent.
- *Engage experts.* Ensure that people with experience in the company's markets, products, and customers are involved in the feature engineering and evolution of the algorithm.

These lessons outline a human-centered approach to machine learning that uses best practices from corporate foresight to engage stakeholders in the process (Figure 1). This type of approach, which integrates human insight at each level of the machine learning stack, will improve the results of the forecast and enhance the ability of the organization to respond meaningfully to the outcomes. At the early stages, ensuring the focus of the project meets known strategic objectives and using experts to brainstorm features will return better initial results. Building a clear engagement plan for refining the algorithm and communicating results to stakeholders regularly can increase both the belief in the forecasts and the organizational will to take action based on the results.

Conclusion

The overall lesson is that machine learning is not an IT project but a business function. Embracing the history of

References

- Columbus, L. 2018. Round-up of machine learning forecasts and market estimates, 2018. *Forbes*, February 18. <https://www.forbes.com/sites/louisacolumbus/2018/02/18/roundup-of-machine-learning-forecasts-and-market-estimates-2018/#9b021a72225c>
- Crews, C. 2015. Killing the official future. *Futures Praxis. Research-Technology Management* 58(3): 59–60.
- De Geus, A. 1997. *The Living Company*. Boston, MA: Harvard Business School Publishing.
- Farrington, T., Henderson, P., and Crews, C. 2012. Research foresights: The use of strategic foresight methods for ideation and portfolio management. *Research-Technology Management* 55(2): 26–33.
- Gonzales, L. 2004. *Deep Survival: Who Lives, Who Dies, and Why*. New York: W.W. Norton and Company.
- Kleiner, A. 2008. *The Age of Heretics*, 2nd edition. San Francisco: Jossey-Bass.
- Knight, W. 2017. The dark secret at the heart of AI. *MIT Technology Review*, April 11. <https://www.technologyreview.com/s/604087/the-dark-secret-at-the-heart-of-ai/>
- Leppert, J., and Kotter, J. 2013. When CEOs talk strategy, 70% of the company doesn't get it. *Forbes*, July 8. <https://www.forbes.com/sites/johnkotter/2013/07/09/heres-why-ceo-strategies-fall-on-deaf-ears/>
- Lorica, B., and Nathan, P. 2018. *The State of Machine Learning Adoption in the Enterprise*. Sebastopol, CA: O'Reilly Media, Inc. <https://www.oreilly.com/data/free/state-of-machine-learning-adoption-in-the-enterprise.csp>
- Van der Heijden, K. 2005. *Scenarios: The Art of Strategic Conversation*. West Sussex, England: Wiley.

Copyright of Research Technology Management is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.